



December 2024

(For BT)

Max. Marks: 100

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

20 Marks

- 1-

12. Functionality of adipic acid is.....
 A) 3 B) 4
 C) 1 D) 2
13. If the arrangement of substituent groups are in alternating fashion in a polymer chain it is calledpolymer.
 A) Syndiotactic B) Atactic
 C) Isotactic D) Graft
14. Which of the following polymer is used in glazing applications?
 A) Teflon B) PMMA
 C) Epoxy resin D) Polyaniline
15. Polymer composites are made of two components, namely.....
 A) Fibre and matrix B) Matrix and Lucite
 C) Fibre and isoprene D) Fibre and Kevlar
16. The combustion of fuel in diesel engines is not instantaneous but there exists an interval of time between fuel injection and start of ignition called.....
 A) Cetane number B) GCV
 C) Ignition lag D) NCV
17. gas produces Carbonic acid
 A) NO₂ B) CO₂
 C) SO₂ D) NO
18. The enzymes which catalyzes oxidation-reduction reaction is.....
 A) Oxidoreductase B) Transferases
 C) Hydrolases D) Ligases
19. Materials wherein two dimensions are at nanoscale range, one dimension is not, is called.....nanomaterials.
 A) Zero dimensional B) One dimensional
 C) Three dimensional D) Two dimensional
20. In approach atoms or molecules are used as building blocks to produce nanomaterials.
 A) Top down B) Crushing
 C) Moulding D) Bottom up

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	What is a reserve battery? Mentions its advantages.	3	L*1	1	1
b)	Describe the construction, working and applications of CH ₃ OH-O ₂ fuel cell.	7	L2	1	1
c)	Explain the construction and working of Photovoltaic cell.	6	L2	1	1
2. a)	Differentiate between temporary and permanent hardness.	4	L3	2	1
b)	Describe the softening of water by hot lime-soda process.	7	L2	2	1
c)	What is Desalination? Describe the desalination of water by reverse osmosis method.	5	L2	2	1
3. a)	Explain the construction, working and applications of Li-ion battery.	6	L2	1	1
b)	Describe the experimental method of determination of total hardness of water by complexometric method.	6	L2	2	1
c)	Calculate the COD of the effluent sample when 25 cm ³ of the effluent requires 9.5 cm ³ of 0.005 M K ₂ Cr ₂ O ₇ for complete oxidation.	4	L3	2	2
Unit – II					
4. a)	Describe the construction of calomel electrode. Explain the experimental method of determination of single electrode potential using calomel electrode.	6	L2	3	1
b)	Explain the different parts of Biosensor with a neat diagram.	5	L2	3	1
c)	Discuss the application of Potentiometry for the redox titration.	5	L2	3	1

5.	a)	Explain the free radical mechanism of addition polymerization taking vinyl chloride as an example.	6	L2	4	1
	b)	Give the synthesis and applications of (i) polyurethane and (ii) silicone rubber.	6	L2	4	1
	c)	What are conducting polymers? Mention any four applications.	4	L1	4	1
6.	a)	Explain the principle and instrumentation of conductometry.	6	L2	3	1
	b)	Explain the synthesis and applications of following polymers (i) Butyl rubber (ii) Epoxy resin	6	L2	4	1
	c)	Describe the mechanism of diesel knocking in IC engine.	4	L2	4	1
Unit – III						
7.	a)	Mention any eight principles of Green Chemistry.	4	L1	5	1
	b)	Describe conventional and green route synthesis of Adipic acid.	6	L2	5	1
	c)	Explain the synthesis of carbon nanotubes by Chemical vapour deposition method.	6	L2	5	1
8.	a)	Give any four advantages of solvent free synthesis. Explain the synthesis of paracetamol by conventional method.	6	L2	5	1
	b)	Properties of materials with nanometer dimensions are significantly different from those of atoms and bulks materials. Why?	2	L1	5	1
	c)	Describe the synthesis of nanomaterials by Sol-gel method.	8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
